Index-calculus leads: status review and missed opportunities

(2026-10-18) **Status.** Desk review across the four repositories (`crypto`, `cryptanalysis`, `crypto-autoresearcher`, `ml-cryptanalysis`) plus a literature scan. No new measurement; every number is quoted from the file named beside it, and each statement is labelled measured, derived, or proposed. No security claim about a deployed curve is made or withdrawn. The previous synthesis is `reports/Index calculus leads next steps.md` (2026-10-14); this note records what changed in the four days since and what the program is still not doing.

Question. Which index-calculus (IC) leads are open today, what did each measure, and which large opportunities is the program missing?

Method. Four parallel sweeps (crypto research directories dated 2026-10-10 to 10-17, backlog cards, problems and journal; the autoresearcher ledger, goals, decisions and open problems; the cryptanalysis and ml-cryptanalysis experiment reports; a 30-query ePrint, arXiv and web scan for 2023–2026), then a cross-check of the pipeline code paths named in the findings.

1. Bottom line

1. **The baseline is unchanged and still trails rho everywhere it is cold.** Table A of `docs/ic/LEADERBOARD.md`: Koblitz one-target IC at 5.23× to 21× the matched rho for $\log_2 r$ 36.6–47.2; Table B prime field 5.47× to 13.5× at 9.7–23.4 bits. The autoresearcher's best complete Koblitz solve is 3.29× [2.80, 4.37] a strengthened rho at N53 (`crypto-autoresearcher/research/cold_arm_composition_ic_20260921/REPORT.md`). No record in any repository claims an end-to-end IC result below rho.
2. **The ten items ranked on 2026-10-14 all ran the same day** (`ic_next10_20261014`), on one shared 4-vCPU, 15 GB container with a 3 GB cap per item. That campaign, plus four improvement rounds and the `dreg_closure_*` series on 10-16/17, closed or narrowed most of the ranked list. The ranked report's statement that all ten are unexecuted is stale.
3. **Two exponent-level questions remain, and both are blocked on memory, not on ideas** (figure 2). The deciding Weil-descent cell (m, ℓ) = (4, 6) needs about 70 GB dense at degree 5 (`dreg_closure_driver_20261016/RESULTS.md`); the PKM tower at m = 3, N = 21 to m = 4, N = 20 needs a 256–512 GB host (`backlog/XP-20261015-pkm-m3-n21-m4-n20`). The instruments that would answer them (msolve, the dense MutantXL closure) already exist and passed their positive controls. **The single largest missed opportunity is one large-memory node for about a week.**
4. **The ECC2K-130 m = 4 price is now bracketed at +10.32 bits over rho in the optimistic case** (degree 5 held to n = 131, 2^71.13 against rho's 2^60.81; figure 1). At that distance the unrun symmetry arms (FGHR 2-torsion, measured 30–416× elsewhere; the equivariant pass) and the unmeasured unit conversion K are no longer rounding error. They matter only if the degree-5 row survives the (4, 6) and $\ell \geq 7$ measurements; until then they are conditional work.
5. **The program's remaining risk is in baselines and bookkeeping**, exactly as the 10-14 report said, and most of that debt is still unpaid: the equal-advice preprocessing-rho control at n = 73/83, the `s4_expr` harness audit, the 48 inherited-F4 replay mismatches, and the column-touch to group-operation conversion K.

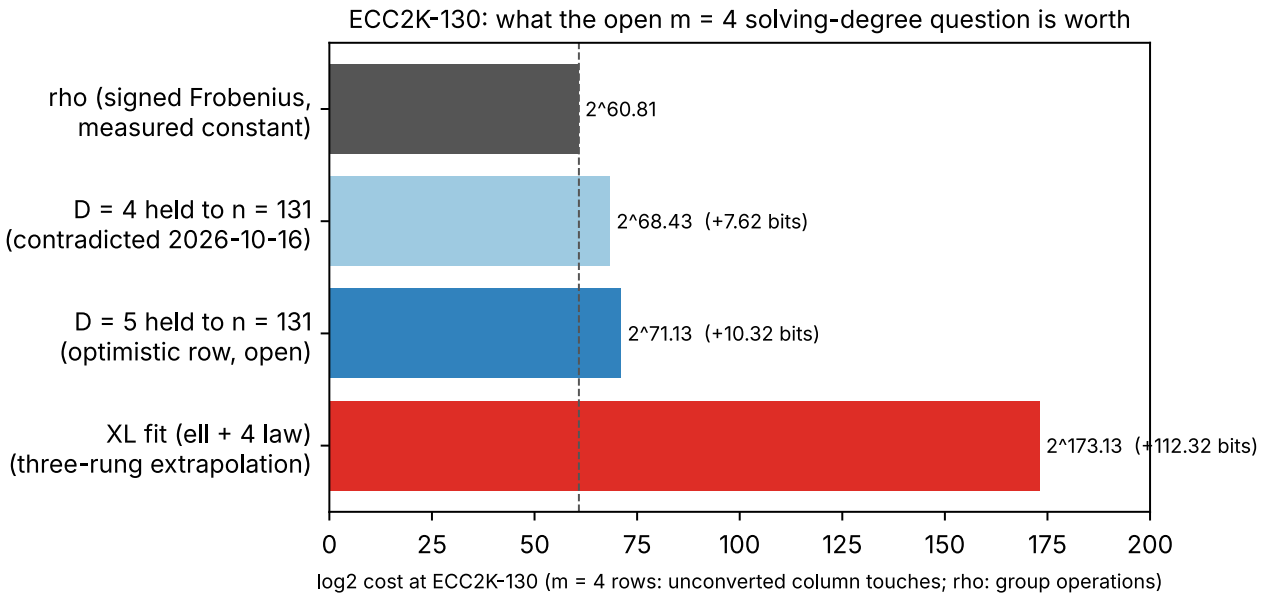


Figure 1. Recorded rows from `research/notes/ecc2k130/RESEARCH_ECC2K130_DESCENT_DEGREE.md` points 6–8. The m = 4 rows are in unconverted column-touch units; rho is in group operations with the measured 2^60.81 constant. Source: `figures/plot.py`.

2. Status of the leads

2.1 The five "do" items of the 2026-10-10 anchor note

#	Lead	Status	What was measured	File
1	Preprocessing-rho control at equal advice (n = 73, 83)	**Open in the specified form.** Ran at n = 41, 53, 61	Bernstein–Lange rho at equal or smaller advice is 53–317× faster online than compact-orbit IC (317× [137, 739] at n = 41; 53× [35, 80] at n = 61, K = 360). The N83 figure (≈50–100×) is a projection: the instrument covers n ≤ 63 and the run did not fit a 1 GiB cap. A later n = 61 pairing over 1024 points (BL 3.37 ms vs IC 111.4 ms) is `claim_check FAIL / SCHEMA_INCOMPLETE` because BL was replayed from an archive, not co-measured. Owner decision 2026-10-15: online rows stay on the board	`ic_next10_20261014/09-equal-advice/README.md`; `ic_improve_20261015/README.md:21`
2	Free relations by pair-root self-collision	**Closed.** Mechanism real, yield not	Homogeneous rank 49 of 599 at n = 61, K = 600; 6 of 599 at n = 83. The anchor note's n^4K^4/r prediction was about 60× too high. Residual constant landed: 0.960× [0.946, 0.976] with free relations on by default	`RESEARCH_FREE_RELATIONS_CENSUS_20261012.md`; `ic_improve_20261015` item 02
3	Solving-degree cells (4,4), (4,5), (4,6) with FGHR and Faugère–Svartz	**Partly measured; deciding cell open.**	msolve F4: (4,4) and (15,5) level at degree 4 where XL read ≥ 7. MutantXL closure refutes 2–3 degrees below XL on 20/20 cells. m = 4 flat at 4 through N = 42; m = 3 rises above 4 at $\ell = 7$ on 8/8 draws. Two survivors: "rise at $\ell = 7$" or "rise once $m \cdot \ell \geq 21$"; (4,6) separates them and is unmeasured (needs ≈70 GB or a sparse closure; the sparse engine built on 10-17 is too slow). **The FGHR arm was not run; no Faugère–Svartz block run exists anywhere.**	`ic_next10_20261014/06-equivariant-f4-m4/README.md`; `dreg_closure_{mutantxl,scale,driver}_20261016/RESULTS.md`; `dreg_closure_specialise_20261017/RESULTS.md`
4	Equivariant F4 to full F4	**Done; not a production lever.**	G2 pass bit-identical to full F_2 RREF; 44–73× was against a naive kernel, 2–3× against `gf2_elim`. Oracle-level verdict TRADES (0.25× at n = 5, 15.5× at n = 13), most of it from fall rows, not equivariance; production never builds σ-symmetric systems, so the hook is dead code. Fall-row gate kept opt-in (`KIC_FALLROW_GATE=1`)	`ic_improve_20261014/02-equivariant-oracle/README.md`; `ic_improve_20261015` item 03
5	D_4-symmetrised S_5 lattice at m = 4	**Lattice route closed generically; the D_4 lattice itself never run.**	Multi-polynomial and shared-target lattices: best S_5 $\delta^* = 0.0140$, 1.12× over 1/80, against $\delta \approx 1/5$ needed; kill criterion met. The D_4 lane instead built a weighted Macaulay solve on $E(F_{\{p^4\}})$: $C_w \approx 1.41 \cdot 10^9$, 844–863× below plain S_4, 18× below C8, handover $n^* =$ 2^118.9; classed engineering	`ic_next10_20261014/05-multipoly-lattice/README.md`; `d4_symmetrised_s5_20261015/README.md`

2.2 The other ranked items and leads opened since 10-10

Lead	Status	Measured	Next step on record
$E(F_{\{p^4\}})$ k = 4 ladder (item 1)	measured, toy	D_4 gives 18.2× [16.4, 20.8] on whole-solve S (constant). Double	stack them; wire sparse LA (§3.3)

		large primes: LA exponent 0.495 ⟶ 0.420 [0.405, 0.440] but relation phase 0.238 ⟶ 0.333 and whole solve 2.31× [2.10, 2.62] costlier, crossing only near 2^219. **The stacked D_4 + 2LP arm was not run.**	
Solution-count census for algebraic bases (item 2)	closed	Kummer, subgroup, Dickson, torus, small-x: 0.93–1.03× decompositions per target	none
PKM tower degree (item 3)	**negative for the m = 4 plateau; m = 3 inconclusive; blocked**	msolve at m = 4, N = 20 completes three productive degree-8 rounds then hits the 5 GB cap: D ≥ 8, about one degree per eight unknowns over N = 12–20. m = 3 at N = 18–24 stopped by the 45-minute wall at D ≥ 7. Positive control 16/16	N = 21–27 on a 256–512 GB host (`pkm_lead1_m3n21_20261016/progress.txt`)
Kummer Fourier spectrum (item 4)	closed	0/31 cells above the null; planted bias detected	none
Cubic GHS ⟶ genus-3 IC (item 7, rounds 2–4)	**largest measured exponent gain**	double large primes q^1.650 ⟶ q^1.300 [1.292, 1.308] (q^1.377 over 2^11–2^15); after tuning the whole solve fell to 0.62–0.65× and the exponent to 1.368 against the 4/3 target; linear algebra (local q^1.52) is the next lever; memory blocks 2^23	LA engine (`ic_improve4_20261015/README.md`)
Quasi-subfield bases (item 8)	closed	yield 0.214 vs 0.204–0.230; trace-zero prefilter never fires on odd-order targets	none
Torelli toy ladder (item 10)	heuristic test	Weil-restriction carriers exist at n = 7	none registered
Binary MITM pair-table oracle ℓ = 8–12	engineering	928× (ℓ = 11), 2,187× (ℓ = 12) over pairs-and-solve, 0 disagreements; relation collection stays Θ(2^n), no vs_rho	ℓ = 13 at ≈1 GiB (`binary_mitm_oracle_l11_l12_20261016/REPORT.typ`)
ECC2K-130 decomposition programme	plan only	best decomposition 42.47 / 55.22 bits above rho; W1–W3 defined, nothing run	`decomposition_programme_20261016/PLAN.md`
IC frontier campaign	phase 1 done	no IC arm beats rho at any band weight and none has a fitted exponent below rho's; phase 2 (RunPod wall + replay) not run	`ic_frontier_campaign_20261015/README.md`
Koblitz linear-row harvest and root-rank gate	landed	1.18–1.89× bench, 1.14–1.20× end to end, defaults	—
Cyclotomic γ-weight base	closed	yield neutral (p = 0.363), cost up to 12% higher	solver clause untested
N83 online / Tier 1 lane	engineering continues	host routes (AWS, GitHub) blocked; A/A noise open	`problems/n83-online-aa-noise.md`
IC probe latency (N83)	frozen, not run	—	`ic_probe_latency_20261017/README.md`

2.3 Autoresearcher ledger (crypto-autoresearcher)

Nineteen IC goals are `active`; almost every one is held by infrastructure, review or custody gates rather than by a negative result. Items that bear on this review:

- **DEC-20261001-9e675e:** enumerative membership is the bottleneck, not IC itself; the approved non-enumerative oracle `EXP-ICEX-153c34` is unbuilt.
- **GOAL-DREG-001 item C1** (n = 15, D = 6): all runs `failed_infrastructure`, peak RSS 11.7–12.2 GiB against an 8 GiB cap. A memory failure recorded as an outcome.

- **H-BINSTD-dcc004** (Krylov/Wiedemann vs dense elimination on IC matrices): approved, never run. **H-ICEX-5666b2** ("the second half of every IC exponent is cited, not measured"): proposed only.
- **H-GFPN-9a29be** (EcGFp5 symmetrisation): inconclusive, red team broke the construction, no m = 5 degree measured. Open, not dead.
- **KN-OPEN-5b3a08** (`harness/semaev.py` `s4_expr` bug, whole-curve sampling) may contaminate 21 archived runs; no audit found.
- Negatives whose ground was process or infrastructure, not measurement: H-STR-002 (missing artifacts), H-DTREE-001 (design infeasible), the ECTD arity-3 panel (17 of 20 arms lost to infrastructure), H-IC-001 (criterion revised post hoc).
- `problems/full-width-ic-arity-grid-unmeasured.md`: no complete 50–63-bit prime or ordinary-binary row exists; msolve refuses the 63-bit field.

2.4 ml-cryptanalysis and cryptanalysis

- **No learned component feeds the measured IC pipeline.** M7/M9 hardness and decomposability predictors are null beyond size and the planted |Aut|/2 law (`experiments/hardness/report.md`, `experiments/fpn-degrees/report.md`). The F_{p^n} degree labels agree with the Macaulay bound on 5,592/5,592 rows.
- cryptanalysis holds the fastest Boolean F4 (`f4_gf2.rs`, GPU: degree-6 matrices 426 s ⟶ 5.7 s on an RTX 5090) and a fitted solving-degree rule D = 2 + [e < −3.5] + [e < −8.5] (MAE 0.06 over 656 bases), with geometric-progression bases 15–40× cheaper per query (`experiments/pdp-degree-heuristics/README.md`). None of these exist in `crypto` under those names, and the rule is not used as a factor-base prior.
- cryptanalysis's own ledger: no engine beats exhaustive search at the PDP stage, the pair table is still the best whole row, and the learned solver selector chose a constant pair-table policy.

2.5 Literature, 2023–2026

The summation-polynomial and Weil-descent literature is dormant since the 2020 subfield-curve (ePrint 2020/1315) and SAT (ePrint 2019/313, WDSat) papers; targeted searches for new solving-degree, tower, Artin–Schreier, Z[τ], sum-product or volcano-decomposition ideas returned nothing. No ECDLP record since 2016–2020; ECC2K-130 remains open. Three external items change how this program should read its own results:

- **Structured generic-group model** (Corrigan-Gibbs, Henzinger, Wu, ePrint 2026/384, EUROCRYPT 2026): DL costs Ω(min(√q, 1/δ)) for algorithms exploiting structure on a δ-fraction of elements, with analogous bounds for elliptic-curve point structure. An EC index calculus must exploit structure in at least a 1/√q fraction of points to beat rho: the decomposition probability wall, stated as a theorem in-model.
- **Non-adaptive time–space bounds** (Dinur, Keller, Marmor, ePrint 2025/783, STOC 2026): with S below √N advice, non-adaptive DL needs T = Ω(√N); only adaptive walks reach ST² ≈ N. Every table-lookup IC hybrid is non-adaptive. This formally closes further pair-table and k-sum engineering at ECC2K-130, independently of the repository's CGK calculation.
- **Crossbred theory** (Vidal, Ionica, Delaplace, ePrint 2024/992; improved preprocessing ePrint 2026/360): current state of the art for overdetermined Boolean systems. In this repository Crossbred was tried only on the m = 2 toy oracle at n = 7, 9 (`research/crossbred_20260920/e2e_panel.txt`), never on the Weil-descent degree cells.

Also surfaced: Galbraith's Õ(q^0.4) volcano navigation (ePrint 2024/924), HATSolver learned Gröbner orderings (arXiv 2512.14722), and the group's own Blackwell rho kernel (ePrint 2026/2469, 14.6–15.9 G iter/s). None narrows the IC-vs-rho gap.

3. Opportunities the program is missing

Ranked by exponent leverage first and cost second. Costs are my estimates from the sizes and timings the notes report, not measurements.

3.1 One large-memory node, one week (the largest)

Both open exponent questions are answered by instruments that already exist and already passed positive controls; what is missing is RAM. Every run since 10-14 has had 3–15 GB (figure 2).

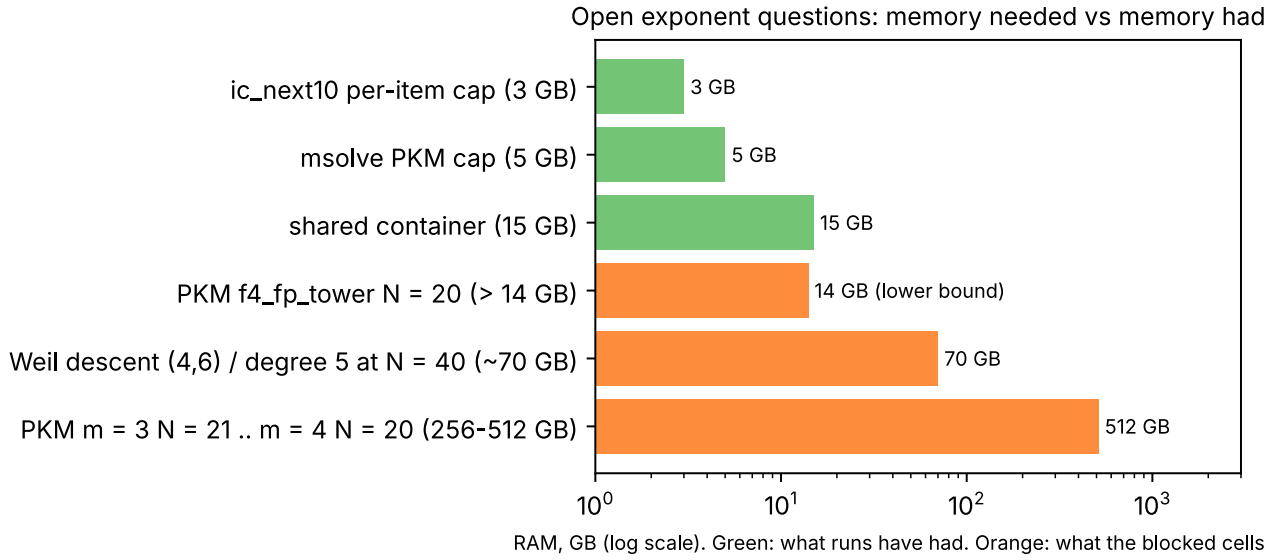


Figure 2. Recorded caps and requirements: `ic_next10_20261014/README.md` and `RULES.md` (3 GB per item, 15 GB container), `03-pkm-degree/README.md` (5 GB msolve cap, f4_fp_tower N = 20 > 14 GB), `dreg_closure_driver_20261016/RESULTS.md` (≈70 GB), `backlog/XP-20261015-pkm-m3-n21-m4-n20` (256–512 GB). Source: `figures/plot.py`.

- **Weil descent, m = 4.** Run (4, 6) with the dense MutantXL closure (≈70 GB at degree 5, N = 40), then m = 3 at ℓ = 8 and m = 4 at ℓ = 7 to measure the slope of the degree past the rise. Three cells decide between "degree 5 holds" (+10.32 bits at ECC2K-130) and "degree follows ℓ" (+112 bits). Nothing else in the binary program has that leverage. Run the FGHR 2-torsion arm in the same session; it was specified on 10-10 and never run.
- **PKM tower, prime fields.** msolve at m = 3, N = 21, 24, 27 and m = 4, N = 20, 24 on 256–512 GB. The m = 4 plateau already broke (D ≥ 8 at N = 20); the m = 3 kill test (D ≥ 8 at N = 21, ≥ 9 at N = 24) could be evaluated neither way under the 45-minute wall. This is the only question in the prime-field line that decides an exponent (`elliptic_snfs_survey_20261016` §7).
- **Autoresearcher DREG C1** at n = 15, D = 6 (needs about 12 GiB) and the (15, 5) degree-8 cell (18–50 GB) ride the same node.

A memory-light alternative worth building only if no node can be had: a GF(2) block Wiedemann (Coppersmith) consistency test on the Macaulay matrix. The repository's block Wiedemann is over Z/r only (`koblitz_sparse_la.rs`); it would give the plain-XL degree, which the closure studies show is 2–3 degrees pessimistic, so it is a bound rather than the measurement.

3.2 Pay the baseline and bookkeeping debt before the next claim

These are cheap, and each one can flip the sign of a headline row.

- **Equal-advice preprocessing rho at n = 73 and 83, co-measured.** Extend the n ≤ 63 instrument to the wide field and run BL and IC in one session under one resource envelope with replay certificates, as the n = 61 diagnostic's own next step says. Record `log₂(S·T²) − log₂(r/|Aut|)` on every online row, as the anchor note asked (not verified to be on the board today).
- **Measure K**, the column-touch to group-operation conversion. Every m = 4 price row at ECC2K-130 is in unconverted units; at +7.6 to +10.3 bits the conversion alone can decide the sign. One perf-ipc session on the current oracle.
- **Audit `s4_expr`** (KN-OPEN-5b3a08) against EXP-MTIC-001 and EXP-FIB-001, and **root-cause the 48 inherited-F4 replay mismatches** (120/168 identical, `ecbench_frontier_20261013`). Inherited F4 is the default Gröbner engine; its counted cost is not reproducible across hosts until this is closed.
- Relabel the autoresearcher negatives whose ground was infrastructure (DREG C1, ECTD panel, H-STR-002, H-DTREE-001) so they stop being cited as results.

3.3 Measure the linear algebra as an object, and make the good

engine reachable The relation matrix has never been measured (REVIEW-UNADDRESSED §4.1 #6; H-ICEX-5666b2). **Correction (2026-10-18, later the same day):** the check below was made on the PR base `f85012683`; `main` already carried `sparse-cado` (CADO-style filter followed by block Wiedemann inside `ic.pipeline`, PR #1874, merge `ae8ac7df9`). What was unreachable from the pipeline was block Lanczos and the unfiltered Krylov cores; PR #2162 wires `wiedemann`, `lanczos`, `filter+lanczos` and `filter+wiedemann` under the full-rank completion rule. Original text: at `f85012683`, ecbench `ic.pipeline` offered only dense or structured Gauss (`src/cryptanalysis/ic_framework/linalg.rs:250-279`; the test

at `:379` asserted that `"wiedemann"` is rejected), while block Wiedemann and block Lanczos with CADO-style filtering sat in `koblitz_sparse_la.rs`. This matters in exactly the two regimes where IC has a sub-rho exponent: the E(F_{p⁴}) ladder (LA is 0.495 of the exponent plain, 0.420 with large primes) and genus-3 after cubic GHS (LA local exponent q^1.52 is the named next lever). H-BINSTD-dcc004 is already approved. Glue work, a few days, and a prerequisite for the first admissible IC bound.

3.4 Stack the measured gains in the proven-exponent regime

- **E(F_{p⁴}): D_4 (18.2× constant) with double large primes (exponent)**. The stacked arm was never run; the plain 2LP arm loses by 2.31× only because its relation phase worsens, which D_4 attacks directly. Run it end to end in ecbench with sparse LA; this is the cheapest route to the first IC row on any frontier.
- **Genus-3 after cubic GHS**: double large primes gave −0.35 in the exponent; the whole solve is at 0.62–0.65×. Add the sparse LA and push past 2^23. Weak class, but it is where the program can build and validate an HPC-grade IC pipeline (filtering, block Wiedemann, replay) before any real target needs one.

3.5 Conditional on the degree: the symmetry arms at m = 4

Only if (4, 6) and the $\ell \geq 7$ cells keep the degree at 5. Then the remaining +10.32 bits is within reach of measured constants that have never been applied to this system: FGHR 2-torsion symmetrisation (30–416× measured in the Gaudry setting, about 5–8.7 bits), the Faugère–Svartz character blocks (never run; the C ≈ D^3.1 projection overstated D_4 by 13×, so discount any projected figure), and the equivariant pass (2–3× honest, plus memory reach). Stack, do not compare; and convert K first (§3.2).

3.6 Literature imports worth a measured trial

- **Crossbred on Weil-descended Semaev systems**, same PDP instances as F4 and WDSat, with the 2026/360 preprocessing. Never tried beyond n = 9, m = 2.
- **Last fall degree** (Huang–Kosters–Yang–Yeo, arXiv 1505.02532) as an order-independent and cheaper instrument than Macaulay refutation for §3.1.
- **Learned orderings** (HATSolver-style) on Semaev and PKM systems: a constant-factor lever, but a cheap one given the ml-cryptanalysis tooling.

3.7 Close the ML and code loops

- Use the cryptanalysis fitted degree rule and the geometric-progression base result as a factor-base prior in `fb-search`; today no learned component touches a measured IC run. Running the ready `fpn_decomposable` config (about 10 CPU-minutes) turns a null into a citable one.
- Port `f4_gf2.rs` / `f4_gpu` (25–75× on degree-6 matrices) into `crypto`, and crypto's newer Koblitz core back into `cryptanalysis`. No GPU IC kernel in `crypto` has ever run on a device.
- Build the non-enumerative membership oracle `EXP-ICEX-153c34`; the ledger's own decision names enumeration as the bottleneck.

3.8 Stop, or label as engineering

Pair-table, k-sum and 6-sum designs at ECC2K-130 and new rungs at n = 85, 89 (now formally non-adaptive, Dinur–Keller–Marmor, on top of the CGK line); first-fall-degree measurements (flat at 3); factor-base shape at m = 2; isogeny, volcano, conductor and Kani transport for Koblitz (h(O_K) = 1, identical ranks; Galbraith 2024/924 lowers navigation cost but there is nothing to navigate to); S_3/S_4 constant-factor work on prime fields; further algebraic factor-base censuses (0.93–1.03×); further Kummer spectra (0/31).

4. Recommended order

#	Action	Regime	Kind	Rough cost
1	(4,6), m = 3 ℓ = 8, m = 4 ℓ = 7 with FGHR arm; PKM N = 21–27	binary and prime	exponent	one 256–512 GB node, about a week
2	Co-measured equal-advice BL rho at n = 73/83; record $S \cdot T^2$ per online row	Koblitz	integrity	1–2 days, needs the wide-field instrument
3	Measure K; audit `s4_expr`; diagnose the 48 replay mismatches	all	integrity	2–4 days
4	Wire block Wiedemann and filtering into `ic.pipeline`; measure the relation matrix	k = 4, genus 3	prerequisite	3–5 days
5	Stacked D_4 + 2LP arm on E(F_{p⁴}), end to end in ecbench	k = 4	exponent (proven regime)	1–2 weeks after 4
6	Crossbred and last-fall-degree on the same PDP cells as 1	binary	instrument	3–5 days

7	Port `f4_gpu` into crypto; build the non-enumerative membership oracle	all	engineering	1–2 weeks
8	Symmetry arms at m = 4 (FGHR, Faugère–Svartz, equivariant)	binary	constant, conditional on 1	2–3 weeks

5. What this review does not establish

It does not show that any IC variant beats rho, cold or online. One statement was corrected after publication: §3.3's "sparse solvers unreachable" was true of the PR base but not of `main`, which already had `sparse-cado` (see the correction there). It does not rerun or re-derive any number; two of the anchor note's own calculations (free-relation count, 60× high; the C ≈ D^3.1 symmetry rule, 13× high) are a reminder that calculations here are not measurements. The costs in §4 are estimates. The sweeps could not see uncommitted worktrees or open PRs, so each action should start with a five-minute check of those places. No canonical graph changes: the leaderboard, scoreboard and progress timeline were checked and carry no new value from this review.

Files

- `README.md` (this report), `report.typ` (generated by Pandoc from it), `report.pdf`.
- `figures/plot.py`, `figures/ecc2k130_degree_rows.{svg,png}`, `figures/memory_wall.{svg,png}`.